Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item

1. Which of the following statements is true? 1 / 1 point

- It's always easiest to verify a recurrence relation using strong induction
- Solving the characteristic equation is the first step in solving a linear recurrence
- The equation defining a recurrence relation is called the Pareto equation

Nice work

Correct. Solving the characteristic equation is the step in solving a linear recurrence.

2. Select from the following the characteristic equation of Fibonacci recurrence relation? 1 / 1 point

- $r^2 - r - 2 = 0$
- $r^2 - r - 1 = 0$
- $r^2 - 2r - 1 = 0$

Nice work

Correct. This is the characteristic equation for the Fibonacci recurrence relation.

3. Which of the following steps do we need to solve the Fibonacci recurrence relation? 1 / 1 point

- Find the multiplicity of the roots
- Build new rules of inference
- Remove all the quantifiers

Nice work

Correct. Finding the multiplicity of the roots is a step in solving the recurrence relation.

4. Which of the following is the characteristic equation of the recurrence relation $a_n = -2a_{n-1} + a_{n-2} - 2a_{n-3}$? 1 / 1 point

- $r^3 + 2r^2 - r + 2 = 0$
- $r^3 + 2r^2 - r - 2 = 0$
- $r^3 - 2r^2 + r - 2 = 0$

Nice work

Correct. This is the characteristic equation for the given recurrence relation.

5. If r is a solution of a characteristic equation with multiplicity p, then which of the following statement is true? 1 / 1 point

- All the combinations of real numbers satisfy the recurrence
- The combination $(\alpha + \beta n + \gamma n^2 + .. + \mu n^{p-1})\boldsymbol{r^n}$ satisfies the recurrence
- The combination $(\alpha + \beta n + \gamma n^2 + .. + \mu n^{p-1})$ satisfies the recurrence

Nice work

Correct. This form accounts for the multiplicity of the root in the solution.